

Isaac Loc

locisaac1223@gmail.com | linkedin.com/in/isaac-loc | github.com/Isaac-Loc

EDUCATION

University at Buffalo

Buffalo, NY

Bachelor of Science in Computer Science, Minor in Math

Expected May 2027

- **Relevant Coursework:** Computer Organization, Algorithms and Complexity, Applied Probability and Statistics, Systems Programming, Data Structures and Algorithms, Object Oriented Programming, Software Engineering

EXPERIENCE

Full Stack Software Engineer

Remote

EZEsports – ezesports.org

Sept 2026 – Present

- Enabled PDF resume uploads with **100% server-side validation** by building the flow in **TypeScript**, storing files in a private **Supabase** bucket, and serving **60-second signed URLs**.
- Engineered on-request erasure of **100% of applicant data**, including resume references, while keeping other records append-only, by extending **PostgreSQL** triggers in a new migration.
- Re-architected applications onto a versioned **JSONB** schema validated on both client and server in TypeScript, removing **~580 lines** of regex parsing and splitting 1 consent into **3 legal checkboxes**.
- Blocked clickjacking, sniffing, and downgrade attacks site-wide by adding **5 HTTP security headers** in **Next.js**, and moved YouTube embeds to a privacy-enhanced, cookie-free mode.
- Refactored a 1,684-line form into a 524-line orchestrator plus **4 section components**, a **~70%** reduction, using **React** and TypeScript with zero visual or accessibility regressions.

Embedded Software Engineer

Buffalo, NY

University at Buffalo

Aug 2025 – Dec 2025

- Synchronized real-time video and image playback across a **50+ LED tile array** by developing **C++** firmware with **ESP-IDF** to coordinate **4 ESP32** microcontrollers and keep every tile in sync.
- Orchestrated live canvas manipulation across **10+ command types** without interrupting active playback, by implementing a C++ command interface backed by a real-time message queue.
- Connected the web frontend to the C++ render loop via **4 REST endpoints**, by building a **Python** canvas library behind a **Flask** API in weekly Agile sprints.

PROJECTS

Project Management Tool | *React, PHP/Slim, MySQL*

GitHub

- Delivered a platform used by **20+ members across 3+ teams**, by architecting a **React** frontend and **PHP/Slim** API with **68 endpoints** over a **17-table MySQL** database.
- Streamlined task tracking with a **6-stage Kanban** board and dependency graph in React, and enabled in-app file access via the **Google Drive API** with OAuth token refresh.

Bitit Auction Marketplace | *React, PHP, MySQL*

- Launched a peer-to-peer marketplace with **2 listing types** (live auctions and buy-it-now), by building a full-stack app with **React**, **PHP**, and **MySQL** for users, listings, and bids.
- Eliminated manual refreshes for auction tracking with **real-time bid updates**, so buyers instantly see the current highest bid, using React and PHP endpoints backed by MySQL.

LEADERSHIP & EXTRACURRICULAR

Brotherhood Outreach & Software Lead | *Theta Tau – Engineering Fraternity*

May 2025 – Present

- Collaborated with the media lead to plan and coordinate **15+ events** for the chapter, achieving a high turnout of **40+ attendees per event** and growing brotherhood and member engagement.
- Built an **LED wall** as software lead to advertise the organization at tabling events, using **C++**, **PlatformIO**, and **ESP32s** driving a **HUB75** LED layout, increasing tabling attraction by **25%**.

TECHNICAL SKILLS

Frameworks & Libraries: React, Next.js, Tailwind CSS, React Aria, Drizzle ORM, PHP/Slim, Flask, ESP-IDF

Databases: PostgreSQL, MySQL, Supabase

Tools & Platforms: Git, Linux, SSH, Vercel, Vitest, CMake, PlatformIO, ESP32, HUB75, REST APIs, OAuth 2.0, Agile/Scrum, Claude Code, AI-Assisted Development

Languages: C/C++, Python, TypeScript, JavaScript, PHP, SQL, HTML/CSS, Java, ARM Assembly